Primal Attack on LWE

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1 Learning with Errors (LWE)

Let an integer modulus $q \geq 2$, a dimension $n \geq 1$ and an error distribution χ over $\mathbb{Z}$. For an $\mathbf{s} \in \mathbb{Z}_q^n$, the LWE distribution $A_{s,\chi}$ over $\mathbb{Z}_q^n \times \mathbb{Z}_q$ is sampled by choosing a uniformly random $\mathbf{a} \in \mathbb{Z}_q^n$ and an error term $e \leftarrow \chi$, outputting $(\mathbf{a}, b = \langle \mathbf{a}, \mathbf{s} \rangle + e \mod q) \in \mathbb{Z}_q^n \times \mathbb{Z}_q$. The distribution χ is necessary to be 0-centered discrte Gaussian distribution with some standard deviation for having the average-case to worst-case reduction[Reg09][Pei09].

The **search** version of the $\mathbf{LWE}_{n,m,q,\chi}$ is, given any desired number of samples $(\mathbf{a}_i, b_i)$ from the LWE distribution $A_{s,\chi}$, one has to find $\mathbf{s}$.

The **decision** version of the $\mathbf{LWE}_{n,m,q,\chi}$ is to distinguish given any desired number of samples $(\mathbf{a}_i, b_i)$ from $A_{\mathbf{s},\chi}$ and the same number of samples drawn from the uniform distribution over $\mathbb{Z}_q^n \times \mathbb{Z}_q$.

Very often, we write these problems in matrix form as follows: collecting the vectors $\mathbf{a}_i \in \mathbb{Z}_q^n$ as columns of a matrix $\mathbf{A} \in \mathbb{Z}_q^{n \times m}$ and the error terms $e_i \in \mathbb{Z}$ and $b_i \in \mathbb{Z}_q$ as the entries of vectors $\mathbf{e} \in \mathbb{Z}^m, \mathbf{b} \in \mathbb{Z}_q^m$ respectively, we have the given inputs

$$\mathbf{A}, \mathbf{b} = \mathbf{A}^\top \mathbf{s} + \mathbf{e} \mod q \tag{1}$$

and asked to find $\mathbf{s}$, or distinguish the input from a uniformly random $(\mathbf{A}, \mathbf{b})$. If $\mathbf{s}$ is chosen uniformly from $\{0, 1\}$ or $\{-1, 0, 1\}$, then this variant is called Binary-LWE. A particularly important fact about the derived encryption schemes in [Reg09, LP10], to reduce the decryption failure, it is essential to hold $|e_i| \ll \lfloor q/4 \rfloor$. Nevertheless, we can redefine LWE as:

$$\mathbf{b} = \mathbf{A}^\top \mathbf{s} + \mathbf{e} \mod q, \mathbf{e} = (e_1, e_2, \ldots, e_m) \text{ and } e_i \leftarrow D_{[\![-a,a]\!],\sigma,0} \tag{2}$$

$[\![-a, a]\!] = \{-a, -a+1, \ldots, 0, \ldots, a-1, a\}$ and $a \ll \frac{q}{4}$ and D is 0-centered discrte Gaussian distribution having support $[\![-a, a]\!]$, standard deviation $\sigma > 0$. One can use the uniform distribution [DMQ13, MP13] as an error distribution over $[\![-a, a]\!]$. We denote the uniform distribution over any set S as $\mathcal{U}(S)$.

2 Hardness of LWE

Theorem 1. *[Reg09] (Informal) Let n and q be integers, $\alpha \in (0, 1)$ be such that $\alpha q > 2\sqrt{n}$, and χ be an error distribution that is assumed to be a discrete Gaussian distribution over $\mathbb{Z}$ centred around 0 with a standard deviation αq. If there exists an efficient algorithm that solves search-LWE, then there exists an efficient quantum algorithm that approximates the GapSVP (decision version of the shortest vector problem) and the SIVP (shortest independent vectors problem) within $\tilde{O}(n/\alpha)$ in the worst case.*

GapSVP and SIVP are two main computational problems on lattices. It is a conjecture that there is no *quantum* polynomial time algorithm that approximates GapSVP or SIVP to within any polynomial factor. The main theorem can be interpreted as suggesting that, based on this conjecture, the Learning With Errors (LWE) problem is difficult to solve. The only evidence supporting this conjecture is the lack of known quantum algorithms for lattice problems that outperform classical algorithms. This absence of evidence is considered to be one of the most significant open questions in the field of quantum computing.

Theorem 2 (Search-to-Decision, Proposition 3 [MM11]). *Let q be positivie integer either prime $q = \Theta(n^c)$ for some constant c or $q = p^e$ for prime $p = \mathrm{poly}(n)$ with distribution $\chi = D_{\mathbb{Z},\sigma}$, $q = p^e = \mathrm{poly}(n)$ with $\chi = \mathcal{U}(\mathbb{Z}_{q^i}), i < e$. Assume there exists a PPT-distinguisher $\mathcal{D}$ that distinguishes the decision version of $\mathbf{LWE}_{n,m,q,\chi}$ with non-negligible advantage, then there exists a PPTalgorithm $\mathcal{A}$ that inverts $\mathbf{LWE}_{n,m,q,\chi}$ i.e. solves the search version of $\mathbf{LWE}_{n,m,q,\chi}$ with non-negligible success-probability.*

So we have search $\mathbf{LWE}_{n,m,q,\chi} \leq$ decision $\mathbf{LWE}_{n,m,q,\chi}$ and decision $\mathbf{LWE}_{n,m,q,\chi} \leq$ search $\mathbf{LWE}_{n,m,q,\chi}$ (obvious!)

Binary-LWE is also hard, as both the papers [BLP+13, MP13] proved. The papers relate (n, q)-Binary-LWE to $(n/O(\log q)), q$-LWE. It is clear that one can solve the Binary-LWE in $O(2^n)$ or $O(3^n)$ operations by trying all the choices for $\mathbf{s}$ and testing whether $\mathbf{b} - \mathbf{As} \pmod q$ is a short vector.

3 LWE as a Lattice problem

Definition 1. ***Bounded Distance Decoding (BDD):*** *Given a lattice basis $\boldsymbol{B}$ and a target vector $\boldsymbol{v}$ such that dist$(\boldsymbol{v}, \Lambda(\boldsymbol{B})) < \lambda_1(\Lambda(\boldsymbol{B}))$, find the lattice vector $\boldsymbol{v} \in \Lambda(\boldsymbol{B})$ closest to $\boldsymbol{v}$.*

Definition 2. ***Unique Shortest Vector (uSVP):*** *Given a lattice $\boldsymbol{B}$ such that $\lambda_2(\Lambda(\boldsymbol{B})) > \lambda_1(\Lambda(\boldsymbol{B}))$, find a nonzero vector $\boldsymbol{v} \in \Lambda(\boldsymbol{B})$ of length $\lambda_1(\Lambda(\boldsymbol{B}))$. $\gamma = \frac{\lambda_2(\Lambda)}{\lambda_1(\Lambda)}$ is defined as Gap.*

Let n, $m > n$, q be positive integers and a matrix $\mathbf{A} \leftarrow \mathcal{U}(\mathbb{Z}_q^{n\times m})$ of rank n. The rows of $\mathbf{A}$ generates $\mathbb{Z}_q^n$ with probability $1 - \frac{1}{p^{m-n-1}}$ where p is the smallest prime factor of q.

We define:

$$\Lambda_q\left(\mathbf{A}^\top\right) = \{\mathbf{v} \in \mathbb{Z}^m : \mathbf{v} \equiv \mathbf{A}^\top\mathbf{s} \mod q \text{ for some } \mathbf{s} \in \mathbb{Z}^n\}$$

This is a q-ary lattice i.e. $q\mathbb{Z}^m \subset \Lambda_q\left(\mathbf{A}^\top\right) \subset \mathbb{Z}^m$.

Note that $\mathbf{A}^\top = \begin{bmatrix}\mathbf{A}_1^\top \\ \mathbf{A}_2^\top\end{bmatrix}$ where $\mathbf{A}_1^\top$ is invertible of order $n \times n$ and $\mathbf{A}_2^\top$ of order $(m-n) \times n$.

Now,

$$\mathbf{v} := \begin{bmatrix}\mathbf{v}_1 \\ \mathbf{v}_2\end{bmatrix} \equiv \mathbf{A}^\top\mathbf{s} \equiv \begin{bmatrix}\mathbf{A}_1^\top \\ \mathbf{A}_2^\top\end{bmatrix}\mathbf{s} \equiv \begin{bmatrix}\mathbf{A}_1^\top\mathbf{s} \\ \mathbf{A}_2^\top\mathbf{s}\end{bmatrix} \mod q$$

So,

$$\begin{aligned} \mathbf{v}_1 \equiv \mathbf{A}_1^\top\mathbf{s} \pmod q \quad \text{and} \quad \mathbf{v}_2 \equiv \mathbf{A}_2^\top\mathbf{s} \pmod q \\ \Longrightarrow \mathbf{v}_2 \equiv \mathbf{A}_2^\top\left(\mathbf{A}_1^\top\right)^{-1}\mathbf{s} \pmod q \\ \Longrightarrow \mathbf{v}_2 = \mathbf{A}_2^\top\left(\mathbf{A}_1^\top\right)^{-1}\mathbf{v}_1 + q\mathbf{u} \quad \text{for some} \quad \mathbf{u} \in \mathbb{Z}^{m-n} \end{aligned}$$

Hence,

$$\mathbf{v} = \begin{bmatrix}\mathbf{v}_1 \\ \mathbf{v}_2\end{bmatrix} = \begin{bmatrix}\mathbf{v}_1 \\ \mathbf{A}_2^\top\left(\mathbf{A}_1^\top\right)^{-1}\mathbf{v}_1 + q\mathbf{u}\end{bmatrix} = \begin{bmatrix}\mathbf{I}_n & \mathbf{0} \\ \mathbf{A}_2^\top\left(\mathbf{A}_1^\top\right)^{-1} & q\mathbf{I}_{m-n}\end{bmatrix}\begin{bmatrix}\mathbf{v}_1 \\ \mathbf{u}\end{bmatrix}$$

So, we can write any $\mathbf{v} \in \Lambda\left(\mathbf{A}^\top\right)$ as $\mathbf{v} = \mathbf{B}\mathbf{w}$ for some $\mathbf{w} \in \mathbb{Z}^m$ where

$$\mathbf{B} = \begin{bmatrix} \mathbf{I}_n & \mathbf{0} \\ \mathbf{A}_2^\top \left(\mathbf{A}_1^\top\right)^{-1} & q\mathbf{I}_{m-n} \end{bmatrix}$$

and $\mathbf{B}$ is an invertible matrix with the determinant q^{m-n}, hence it is a basis of the lattice $\Lambda_q\left(\mathbf{A}^\top\right)$.

Now, if $\mathbf{e}$ is small $\left(||\mathbf{e}||_2 < \frac{\lambda_1\left(\Lambda_q\left(\mathbf{A}^\top\right)\right)}{2}\right)$, the $\mathbf{LWE}_{n,m,q,\chi}$ can be seen as follows: given a point $\mathbf{b}$ as a target vector, one has to find the lattice point $\mathbf{v}$ in $\Lambda_q\left(\mathbf{A}^\top\right)$ so that $\mathbf{e} = \mathbf{b} - \mathbf{v} = \mathbf{b} - \mathbf{B}\mathbf{u}$ for some $\mathbf{u} \in \mathbb{Z}^m$. So we have reduced $\mathbf{LWE}_{n,m,q,\chi}$ to a BDD problem on the lattice $\Lambda_q\left(\mathbf{A}^\top\right)$. Now we will embed $\Lambda_q\left(\mathbf{A}^\top\right)$ into an $m+1$ dimensional lattice $\Lambda = \Lambda(\mathbf{B}')$ where

$$\mathbf{B}' = \begin{bmatrix} \mathbf{B} & \mathbf{b} \\ \mathbf{0} & \mu \end{bmatrix}.$$

Here, μ is called the embedding constant, and the best choice is $\mu = ||\mathbf{e}||_2$ (by Theorem 1 of [LM09]). Note that:

$$\begin{bmatrix} \mathbf{e} \\ \mu \end{bmatrix} = \begin{bmatrix} -\mathbf{B}\mathbf{u} + \mathbf{b} \\ \mu \end{bmatrix} = \mathbf{B}' \begin{bmatrix} -\mathbf{u} \\ 1 \end{bmatrix}$$

We will show that if $\mu = ||\mathbf{e}||_2$ and $\left(||\mathbf{e}||_2 < \frac{\lambda_1\left(\Lambda_q\left(\mathbf{A}^\top\right)\right)}{2}\right)$, then $\Lambda(\mathbf{B}')$ will contain a unique shortest vector $\mathbf{v}' = \begin{bmatrix} \mathbf{e} \\ \mu \end{bmatrix}$. Thus, finding such a vector will solve the BDD problem and so the $\mathbf{LWE}_{n,m,q,\chi}$. $||\mathbf{v}'||_2 = \sqrt{\mu^2 + \mu^2} = \sqrt{2}\mu$. Our claim is that all the other vectors in $\Lambda(\mathbf{B}')$ that are not linearly dependent to $\mathbf{v}'$ is of length at least $\frac{\lambda_1\left(\Lambda_q\left(\mathbf{A}^\top\right)\right)}{\sqrt{2}}$. We will prove our claim by contradiction. Let there exist a vector $\mathbf{w}' \in \Lambda(\mathbf{B}')$ such that $||\mathbf{w}'||_2 < \frac{\lambda_1\left(\Lambda_q\left(\mathbf{A}^\top\right)\right)}{\sqrt{2}}$ that is linearly independent of $\mathbf{v}'$. Rewrite $\mathbf{w}' = \begin{bmatrix} \mathbf{B} & \mathbf{b} \\ \mathbf{0} & \mu \end{bmatrix} \begin{bmatrix} \mathbf{x} \\ y \end{bmatrix} = \begin{bmatrix} \mathbf{B}\mathbf{x} + y\mathbf{b} \\ y\mu \end{bmatrix} = \begin{bmatrix} \mathbf{w} + y\mathbf{b} \\ y\mu \end{bmatrix}$, y is an negative integer and $\mathbf{w} = (\mathbf{B}\mathbf{x}) \in \Lambda_q\left(\mathbf{A}^\top\right)$.

$$\|\mathbf{w}'\|_2 = \sqrt{\|\mathbf{w} + y\mathbf{b}\|_2^2 + (y\mu)^2} < \frac{\lambda_1\left(\Lambda_q\left(\mathbf{A}^\top\right)\right)}{\sqrt{2}}$$

$$\implies y\mu < \frac{\lambda_1\left(\Lambda_q\left(\mathbf{A}^\top\right)\right)}{\sqrt{2}} \quad \text{and} \quad \|\mathbf{w} + y\mathbf{b}\|_2 < \sqrt{\frac{\lambda_1\left(\Lambda_q\left(\mathbf{A}^\top\right)\right)^2}{2} - (y\mu)^2}.$$

Consider $\mathbf{w} + y\mathbf{v} \in \Lambda_q\left(\mathbf{A}^\top\right)$ and we have assumed that $\mathbf{w}'$ is linearly independent of $\mathbf{v}'$, so $\mathbf{w} + y\mathbf{v}$ is a non-zero lattice vector(why?). To get the contradiction, we will show that the length of the vector $\mathbf{w} + y\mathbf{v}$ is strictly less than $\lambda_1\left(\Lambda_q\left(\mathbf{A}^\top\right)\right)$.

$$||\mathbf{w} + y\mathbf{v}||_2 = ||\mathbf{w} + y\mathbf{b} + y(\mathbf{b} - \mathbf{v})||_2 \leq ||\mathbf{w} + y\mathbf{b}||_2 + y||\mathbf{b} - \mathbf{v}||_2 \leq \sqrt{\frac{\lambda_1\left(\Lambda_q\left(\mathbf{A}^\top\right)\right)^2}{2} - (y\mu)^2} + y\mu.$$

The above inequality is maximized when $y = \frac{\lambda_1\left(\Lambda_q\left(\mathbf{A}^\top\right)\right)}{2}$ and therefore for all y,

$$||\mathbf{w} + y\mathbf{v}||_2 < \sqrt{\frac{\lambda_1\left(\Lambda_q\left(\mathbf{A}^\top\right)\right)^2}{2} - (y\mu)^2} + y\mu \leq \lambda_1\left(\Lambda_q\left(\mathbf{A}^\top\right)\right),$$

which gives the contradiction.

4 Uniqueness of the Solution

It is a trivial question whether LWE consists of a unique solution. To be more precise, under what conditions on the parameters m and n, is there only one solution to an LWE problem? As we saw in the previous section, LWE can be reduced to a BDD problem on the lattice $\Lambda_q\left(\mathbf{A}^\top\right)$. This is exactly the same as the "Decoding Problem" in coding theory. The decoding problem refers to the challenge of recovering the original message (or code word) from a received message that may have been corrupted during transmission over a noisy channel. So the question is under what conditions on the m and n and the error $\mathbf{e}$, can we recover the solution of the LWE? The $||\mathbf{e}||_2$ has to be strictly lesser than $\frac{\lambda_1\left(\Lambda_q\left(\mathbf{A}^\top\right)\right)}{2}$, otherwise the BDD problem has no unique solution. Next, if we assume that the $\text{LWE}_{m,n,q,\chi}$ has an unique solution, and already we have the condition $||\mathbf{e}||_2 < \frac{\lambda_1\left(\Lambda_q\left(\mathbf{A}^\top\right)\right)}{2}$, then by the Gaussian Heuristic: $\lambda_1\left(\Lambda_q\left(\mathbf{A}^\top\right)\right) \approx \sqrt{\frac{m}{2\pi e}}\,\text{Vol}\left(\Lambda_q\left(\mathbf{A}^\top\right)\right)^{\frac{1}{m}} = \sqrt{\frac{m}{2\pi e}}q^{\frac{m-n}{m}}$ and since the error distribution χ is taken to be discrete Gaussian, then expected norm of the error vector $\mathbf{e}$ is $\approx \sqrt{m}\alpha q$, we have :

$$\sqrt{m}\alpha q < \frac{\sqrt{\frac{m}{2\pi e}}q^{\frac{m-n}{m}}}{2} \implies m > k \times n \times \log q, k = \frac{1}{\log \frac{1}{2\alpha\sqrt{2\pi e}}} > 0 \tag{3}$$

where k is a positive real number. So to keep a unique solution of an $\text{LWE}_{m,n,q,\chi}$, we have to set the values of m, n, q, α so that Eq. 3 holds.

5 The Primal Attack

The primal attack on the Learning With Errors (LWE) problem was first formally introduced in the [ADPS16]. Specifically, the technique is discussed in the NewHope paper, where it is outlined: reducing LWE to a unique Shortest Vector Problem (uSVP) via lattice embedding, then applying lattice reduction algorithms to recover the secret. We also discussed a lattice-embedding technique, called Kannan's embedding, which is the first form of the so-called Primal Attack. We will first see how many LWE samples are required to perform the primal attack via Kannan's embedding. Gama and Nguyen [GN08] have proposed a heuristic for estimating the performance of lattice basis reduction algorithms. Let us consider a lattice basis reduction algorithm that takes as input a lattice L of dimension m and outputs a list of vectors $\mathbf{b}_1, \mathbf{b}_2, \ldots, \mathbf{b}_m$. The root Hermite factor $\delta \in \mathbb{R}^+$ of a lattice basis reduction algorithm is:

$$\delta = \frac{||\mathbf{b}_1||_2^{\frac{1}{m}}}{\text{Vol}(\Lambda)^{\frac{1}{m^2}}} \tag{4}$$

The paper [GN08] argues that $\delta = 1.01$ is about the limit of the practical algorithm BKZ. The paper [CN11] extended the study to algorithms with greater running time. Their heuristic argument is that $\delta = 1.006$ could be achieved by an algorithm performing around 2^{110} operations. In section 3.3 of [GN08], the authors drew attention to the unique-SVP problem. If one knows that there is a large gap γ, then a lattice basis reduction can solve the unique-SVP with some Hermite factor δ. Gama and Nguyen observed that the practical algorithms succeed if $\gamma > c\delta^m$ for some small constant $c < 1$. This Gama-Nguyen heuristic is called "**Estimate 2008**". The lower the value of δ, the lattice basis reduction algorithm needs to perform more operations. The root hermite factor δ

is related to the BKZ block size β [Che13] by:

$$\delta(\beta) = \left(\frac{\beta}{2\pi e} \cdot (\pi\beta)^{1/\beta}\right)^{\frac{1}{2(\beta-1)}}, \quad \beta \geq 50 \tag{5}$$

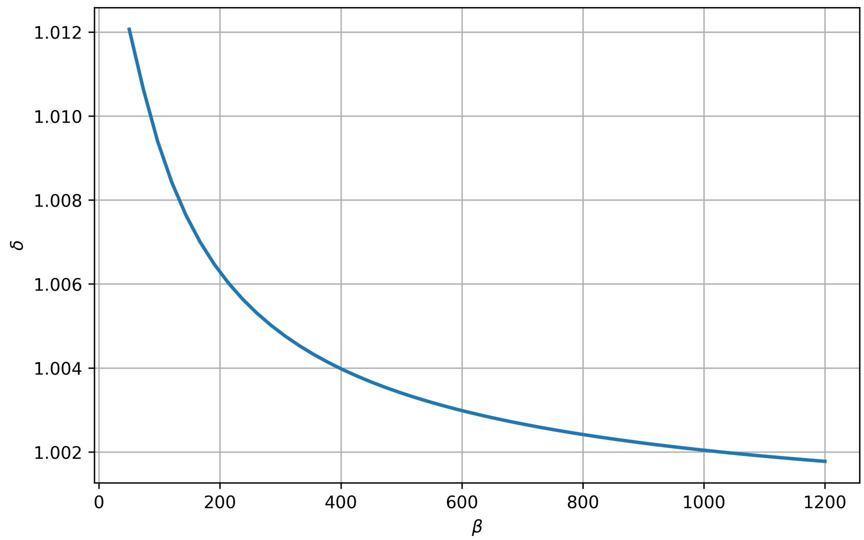


Figure 1: Graph of $\delta(\beta)$ for $50 \leq \beta \leq 1200$

Now we will see how "**Estimate 2008**" will help us to determine the sample size of LWE to perform a standard attack. Consider running the embedding technique on an LWE instance, using the lattice Λ given by the matrix $\mathbf{B}'$. We will have a good chance of getting the right result if the error vector $\mathbf{e}$ is very short compared with the second shortest vector in the lattice Λ, which we assume to be the shortest vector in the original lattice Λ_q. We know by the Gaussian heuristic that the Euclidean length of the shortest vector in the lattice Λ_q is approximately $\sqrt{\frac{m}{2\pi e}} q^{\frac{m-n}{m}}$. But Λ_q also has some known vectors, called "q-vectors," of Euclidean length q. Hence we have

$$\lambda_2(\Lambda(\mathbf{B}')) \approx \lambda_1(\Lambda_q(\mathbf{B})) \approx \min\left(q, \sqrt{\frac{m}{2\pi e}} q^{\frac{m-n}{m}}\right).$$

On the other hand, the vector $\mathbf{e}$ has expected euclidean length is $\sqrt{m}\alpha q$ as components of $\mathbf{e}$ are from Gaussian over $\mathbb{Z}$ with standard deviation αq and so the expected length of the vector $\begin{bmatrix} \mathbf{e} \\ \mu \end{bmatrix}$ is $\sqrt{2m}\alpha q$ when $\mu = \sqrt{m}\alpha q$. Assume that $\lambda_1(\Lambda(\mathbf{B}')) \approx \sqrt{2m}\alpha q$. Hence the gap:

$$\gamma(m) = \frac{\lambda_2(\Lambda(\mathbf{B}'))}{\lambda_1(\Lambda(\mathbf{B}'))} \approx \frac{\min\left(q, \sqrt{\frac{m}{2\pi e}} q^{\frac{m-n}{m}}\right)}{\sqrt{2m}\alpha q}$$

For a successful attack, we want this gap γ to be large, so we need:

$$\sqrt{2m}\alpha q \ll \sqrt{\frac{m}{2\pi e}} q^{\frac{m-n}{m}} < q$$

To determine whether an LWE instance can be solved using the embedding technique and a lattice reduction algorithm with a given root Hermite factor δ, one chooses a sample size m and verifies the gap condition $\gamma(m) > c\delta^{m+1}$ for a suitable value c. Since c is unknown so we can maximize the function f for fixed n, q, δ to determine the optimal m, $f(m) = \frac{q^{-\frac{n}{m}}}{2\sqrt{\pi e}\alpha\delta^{m+1}}$. Taking the log of both sides and the first derivative test tells us

$$\frac{n}{m^2}\ \log q = \log\delta \implies m = \sqrt{\frac{n\ \log q}{\log\delta}}$$

One can check that at $m = \sqrt{\frac{n\ \log q}{\log\delta}}$, f attains its global maxima.

Parameters	1	2	3	4	5	6	7	8	9	10
n	128	256	512	1024	1280	1536	1624	1792	1824	2048
δ	1.007200	1.006444	1.005689	1.004933	1.004178	1.003422	1.002667	1.001911	1.001156	1.000400
m	534	798	1200	1822	2213	2678	3118	3869	5018	9036

Table 1: Old Attack's Parameters n, δ, m, $q = 8380417$

5.1 A New Attack due to Bai-Galbraith Embedding

Now, we will discuss another embedding technique that yields a smaller sample size than the prior. Bai and Galbraith [BG14] introduced a new attack on Binary-LWE. They used the fact that $\mathbf{s}$ is short. The previous attack on LWE can't use this shortness of $\mathbf{s}$. Let us have an LWE instance $(\mathbf{A}^{\top}_{m\times n}, \mathbf{b})$.
Now write,

$$\mathbf{b} = \mathbf{A}^{\top}\mathbf{s} + \mathbf{e} \pmod q \implies \mathbf{b} = [\mathbf{A}^{\top}|\mathbf{I}_m]\begin{bmatrix}\mathbf{s}\\ \mathbf{e}\end{bmatrix} \pmod q \tag{6}$$

In 6, we have just converted an LWE instance to an ISIS instance. $\begin{bmatrix}\mathbf{s} \mid \mathbf{e}\end{bmatrix}^{\top}$ is our short vector. Now, we will reduce this ISIS problem to a closest-vector problem.
Let $\mathbf{w} = \begin{bmatrix}\mathbf{0}\\ \mathbf{b}\end{bmatrix}$ be a target vector and consider the lattice $L = \{\mathbf{v} \in \mathbb{Z}^{m+n} : [\mathbf{A}^{\top}|\mathbf{I}_m]\mathbf{v} \equiv \mathbf{0}\ mod\ q\}$. To solve the CVP instance $(L, \mathbf{w})$, we have to construct a basis of the lattice L. A basis can be constructed as:

$$\begin{aligned}
&[\mathbf{A}^{\top} \mid \mathbf{I}_m]\mathbf{v} \equiv 0 \pmod q\\
\implies &[\mathbf{A}^{\top} \mid \mathbf{I}_m]\mathbf{v} = q\mathbf{u}, \quad \mathbf{u} \in \mathbb{Z}^m\\
\implies &\mathbf{A}^{\top}\mathbf{v}_1 + \mathbf{v}_2 = q\mathbf{u}, \quad \text{where } \mathbf{v} = \begin{bmatrix}\mathbf{v}_1\\ \mathbf{v}_2\end{bmatrix}\\
\implies &\mathbf{v}_2 = q\mathbf{u} - \mathbf{A}^{\top}\mathbf{v}_1\\
\implies &\mathbf{v} = \begin{bmatrix}\mathbf{v}_1\\ \mathbf{v}_2\end{bmatrix} = \begin{bmatrix}\mathbf{v}_1\\ q\mathbf{u} - \mathbf{A}^{\top}\mathbf{v}_1\end{bmatrix}\\
&= \begin{bmatrix}\mathbf{I}_n & \mathbf{0}\\ -\mathbf{A}^{\top} & q\mathbf{I}_m\end{bmatrix}\begin{bmatrix}\mathbf{v}_1\\ \mathbf{u}\end{bmatrix}.
\end{aligned}$$

So, $\mathbf{B} = \begin{bmatrix}\mathbf{I}_n & \mathbf{0}\\ -\mathbf{A}^{\top} & q\mathbf{I}_m\end{bmatrix}$ is a generating matrix and has determinant q^m. So $\mathbf{B}$ is a basis of L. Now to solve CVP instance $(L, \mathbf{w})$, one seeks a vector $\mathbf{v} \in L$ so that $\mathbf{v} = \mathbf{Bz}$ for some $\mathbf{z} \in \mathbb{Z}^{m+n}$. So it is expected that $\mathbf{w} - \mathbf{v} = \begin{bmatrix}\mathbf{s} \mid \mathbf{e}\end{bmatrix}^\top$ and $\mathbf{v} = \begin{bmatrix} -\mathbf{s} \\ \mathbf{b} - \mathbf{e} \end{bmatrix}$. Here, the main observation is CVP finds an unbalanced solution $\begin{bmatrix}\mathbf{s} \mid \mathbf{e}\end{bmatrix}^\top$ because of $||\mathbf{s}||_2 < ||\mathbf{e}||_2$. Assume that $\mathbf{s} \xleftarrow{\$} \{-1, 0, 1\}^n$. So to get a balanced solution (i.e. $\alpha q||\mathbf{s}||_2 \approx ||\mathbf{e}||_2$) $\begin{bmatrix} -\alpha q \mathbf{s} \\ \mathbf{e} \end{bmatrix} = \mathbf{w} - \mathbf{v}$, one has to multiply the first n rows of $\mathbf{B}$ with αq to seek $\mathbf{v} = \begin{bmatrix} -\alpha q \mathbf{s} \\ \mathbf{b} - \mathbf{e} \end{bmatrix} = \begin{bmatrix} \alpha q \mathbf{I}_n & \mathbf{0} \\ -\mathbf{A}^\top & q\mathbf{I}_m \end{bmatrix} \begin{bmatrix} -\mathbf{s} \\ \mathbf{0} \end{bmatrix}$. When $\mathbf{s} \xleftarrow{\$} \{0, 1\}^n$, then to balance the solution, so that the components of $\mathbf{s}$ will be symmetric about 0 (i.e. each component of $\mathbf{s}$ will be either $-\alpha q$ or $+\alpha q$),

1. update the target vector from $\mathbf{w} = \begin{bmatrix} \mathbf{0} \\ \mathbf{b} \end{bmatrix}$ to $\mathbf{w}_{\text{updated}} := \mathbf{w} - (\alpha q, \alpha q, \ldots, \alpha q, 0, \ldots, 0)^\top$
2. multiply first n rows of $\mathbf{B}$ with $2\alpha q$.

We want a balanced solution: $(\pm\alpha q, \pm\alpha q, \ldots, \pm\alpha q, e_1, \ldots, e_m)^\top = \mathbf{w}_{\text{updated}} - \mathbf{v}$. To get this, one seeks $\mathbf{v} = \begin{bmatrix} 2\alpha q \mathbf{I}_n & \mathbf{0} \\ -\mathbf{A}^\top & q\mathbf{I}_m \end{bmatrix} \begin{bmatrix} -\mathbf{s} \\ \mathbf{0} \end{bmatrix} = \begin{bmatrix} -\mathbf{2}\alpha q \mathbf{s} \\ \mathbf{A}^\top \mathbf{s} \end{bmatrix} = \begin{bmatrix} -\mathbf{2}\alpha q \mathbf{s} \\ \mathbf{b} - \mathbf{e} \end{bmatrix}$. The vector $-2\alpha q\mathbf{s}$ has each component is either 0 or $-2\alpha q$. When we subtract $\mathbf{v}$ from $\mathbf{w}_{\text{updated}}$ then the first n components of the resultant vector will be either $-\alpha q$ or $+\alpha q$ and rest respective m components are $e_1, \ldots, e_m$. By re-balancing, the volume of the lattice got increased by the factor $(\alpha q)^n$ for the case $\{-1, 0, 1\}$ and $(2\alpha q)^n$ for the case $\{0, 1\}$. So it is expected that the *gap* in the re-scaled lattice is larger than compared in the original lattice. The basis of the embedded lattice, for the $\mathbf{s} \in \{-1, 0, 1\}^n$, is $\mathbf{B}' = \begin{bmatrix} \alpha q \mathbf{I}_n & \mathbf{0} & \mathbf{0} \\ -\mathbf{A}^\top & q\mathbf{I}_m & \mathbf{b} \\ \mathbf{0} & \mathbf{0} & 1 \end{bmatrix}$. Consider the lattice $L' = \{\mathbf{x} \in (\alpha q\mathbb{Z})^n \times \mathbb{Z}^{m+1} : \left(\frac{1}{\alpha q}\mathbf{A}^\top |\mathbf{I}_m| - \mathbf{b}\right) \mathbf{x} \equiv \mathbf{0} \ (\ mod\ q)\}$ which has the basis $\mathbf{B}'$. So we have the explicit form of the embedded lattice. Note that $\mathbf{u} = \begin{bmatrix}\alpha q\mathbf{s} \mid \mathbf{e} \mid 1\end{bmatrix}^\top \in L'$. To expect $\mathbf{u}$ is the unique shortest vector in L', it is essential to keep

$$\alpha q\sqrt{m+n} < \sqrt{\frac{m+n}{2\pi e}} \left(q^m (\alpha q)^n\right)^{\frac{1}{m+n}} \implies \alpha^{\frac{m}{m+n}} < \frac{1}{\sqrt{2\pi e}} \tag{7}$$

So assuming 7 and we have $\lambda_1(L') \approx \alpha q\sqrt{m+n}$ and $\lambda_2(L') \approx \sqrt{\frac{m+n}{2\pi e}} \left(q^m (\alpha q)^n\right)^{\frac{1}{m+n}}$. To determine the optimal m, consider $f(m) = \frac{\lambda_2(L')}{\lambda_1(L')\cdot\delta^{m+n+1}} = \frac{\alpha^{\frac{-m}{m+n}} \sqrt{\frac{m+n}{2\pi e}}}{\sqrt{m+n}\cdot\delta^{m+n+1}} = \frac{\alpha^{\frac{-m}{m+n}}}{\sqrt{2\pi e}\cdot\delta^{m+n+1}}$. Taking the log of both sides and the first derivative test tells us that m attains the max value:

$$-\frac{n}{(m+n)^2} \log\alpha - \log\delta = 0 \implies m = \sqrt{\frac{n(\log q - \log\sigma)}{\log\delta}} - n, \sigma := \alpha q$$

Parameters	1	2	3	4	5	6	7	8	9	10
n	128	256	512	1024	1280	1536	1624	1792	1824	2048
δ	1.004100	1.003689	1.003278	1.002867	1.002456	1.002044	1.001633	1.001222	1.000811	1.000400
m	505	675	866	1029	1188	1415	1766	2316	3261	5605

Table 2: New Attack's Parameters n, δ, $\alpha q = 2\sqrt{n}$, and m values for $q = 8380417$.

From the Graph-1, Table-1 and Table-2, one can observe that for the dimension $n = 1024$ the required size of LWE samples is reduced from 1822 to 1029 in the new attack, and to recover the unique shortest vector, the block size β is needed around $600 - 650$. The cost of BKZ reduction is dominated by its shortest vector problem (SVP) oracle, which in the state-of-the-art setting is based on lattice sieving. The best known classical sieving algorithms [BDGL16] solve SVP in time approximately $2^{0.292\beta}$ and space $2^{0.207\beta}$, where β is the BKZ block size. Since bit security is defined as the base-2 logarithm of the attack cost, the security level against classical attacks is estimated as about 0.292β bits. For example, block sizes of $\beta = 300, 400$, and 500 correspond to roughly $(\log_2(2^{0.292\beta}) = 0.292\beta) = 88, 117$, and 146 bits of classical security, respectively. In the quantum setting, Grover-accelerated sieving [Laa16] improves the running time to $2^{0.265\beta}$, yielding an estimated 0.265β bits of security. Polynomial overheads in the lattice dimension contribute only negligible factors, so security analyses typically rely on these linear approximations when relating β to bit security.

5.2 Short-Secret LWE

Recall the definition 2 and assume that errors are coming from $\mathcal{U}[\![-a, a]\!]$. According to [ACPS09, MR09, LP10, BCD$^+$16] secret and error, both can be chosen from the same distribution.

Definition 3. *Let m,n and q be positive integers, $\boldsymbol{s} \leftarrow \mathcal{U}[\![-a, a]\!]^n$ and $\boldsymbol{e} \leftarrow \mathcal{U}[\![-a, a]\!]^m$ where $d \ll \frac{q}{4}$. Given $\boldsymbol{A} \in \mathbb{Z}_q^{n\times m}$ and $\boldsymbol{b} = \boldsymbol{A}^\top \boldsymbol{s} + \boldsymbol{e} \mod q$, one has to find $\boldsymbol{s}$, or distinguish the input from a uniformly random $(\boldsymbol{A}, \boldsymbol{b})$.*

The LWE in the definition 3 is called the Short-Secret LWE and we denote it by $\mathbf{LWE}_{m,n,q,a}$.

Lemma 1. *search* $\mathbf{LWE}_{m,n,q,a} \leq$ *search* $\mathbf{LWE}_{n,m,q,\chi}$ *with* $\chi = \mathcal{U}[\![-a, a]\!]$

Proof. Let $\mathbf{b} = \mathbf{A}^\top \mathbf{s} + \mathbf{e} \mod q$ be a $\mathbf{LWE}_{m,n,q,a}$ instance and $\mathbf{d} \in \mathbb{Z}_q^n$. Now $\mathbf{b}' = \mathbf{b} + \mathbf{A}^\top \mathbf{d} \mod q \implies \mathbf{b}' = \mathbf{A}^\top \mathbf{s} + \mathbf{e} + \mathbf{A}^\top \mathbf{d} \mod q \implies \mathbf{b}' = \mathbf{A}^\top(\mathbf{s} + \mathbf{d}) + \mathbf{e}$, where $\mathbf{s} + \mathbf{d} \in \mathbb{Z}_q^n$. Then $(\mathbf{A}, \mathbf{b}')$ is an $\mathbf{LWE}_{n,m,q,\chi}$ instance with $\chi = \mathcal{U}[\![-a, a]\!]$. ◻

Lemma 2. *search* $\mathbf{LWE}_{n,m,q,\chi} \leq$ *search* $\mathbf{LWE}_{m-n,n,q,a}$ *with* $\chi = \mathcal{U}[\![-a, a]\!]$

Proof. Let $\mathbf{b} = \mathbf{A}^\top \mathbf{s} + \mathbf{e} \mod q$ be a $\mathbf{LWE}_{n,m,q,\chi}$ instance with $\chi = \mathcal{U}[\![-a, a]\!]$.

$$\mathbf{b} = \begin{bmatrix} \mathbf{b}_1 \\ \mathbf{b}_2 \end{bmatrix} = \begin{bmatrix} \mathbf{A}_1^\top \\ \mathbf{A}_2^\top \end{bmatrix} \mathbf{s} + \begin{bmatrix} \mathbf{e}_1 \\ \mathbf{e}_2 \end{bmatrix}$$

where $\mathbf{A}_1^\top$ is invertible matrix in $\mathbb{Z}_q$ with high probability of order n. Let $\bar{\mathbf{A}} = -\mathbf{A}_2^\top (\mathbf{A}_1^\top)^{-1}$. Then $\bar{\mathbf{b}} = \bar{\mathbf{A}}\mathbf{b}_1 + \mathbf{b}_2 = \bar{\mathbf{A}}\mathbf{e}_1 + \mathbf{e}_2$. Thus $(\bar{\mathbf{A}}, \bar{\mathbf{b}})$ is an search $\mathbf{LWE}_{m-n,n,q,a}$ instance. ◻

Theorem 3. *search short-secret LWE and search LWE are equivalent.*

Proof. By combining Lemma 1 and Lemma 2, we can prove it. ◻

5.3 Orthogonal Projection and Projected Lattice

5.3.1 Orthogonal Projection.

Definition 4. *Let V be an inner product space, W be any subset of V. We define $W^\perp := \{x \in V : \langle x, y\rangle = 0; \forall y \in W\}$ and $W^\perp$ is called orthogonal complement of W. If W is a subspace of V then $W^\perp$ is also a subspace of V.*

Theorem 4. *Let V be a d-dimensional inner product space, W be any subspace of V and $\boldsymbol{y} \in V$. Then $\exists$ unique vectors $\boldsymbol{u} \in W$ and $\boldsymbol{z} \in W^{\perp}$ such that $\boldsymbol{y} = \boldsymbol{u} + \boldsymbol{z}$. Moreover if $\{\boldsymbol{b}_1^*, \boldsymbol{b}_2^*, \ldots, \boldsymbol{b}_k^*\}$ is an orthogonal basis for W, then $\{\boldsymbol{b}_1^*, \boldsymbol{b}_2^*, \ldots, \boldsymbol{b}_k^*\}$ can be extended to an orthogonal basis $\{\boldsymbol{b}_1^*, \boldsymbol{b}_2^*, \ldots, \boldsymbol{b}_k^*, \boldsymbol{b}_{k+1}^*, \ldots, \boldsymbol{b}_d^*\}$ for V and $\{\boldsymbol{b}_{k+1}^*, \boldsymbol{b}_{k+2}^*, \ldots, \boldsymbol{b}_d^*\}$ is an orthogonal basis of $W^{\perp}$.*

Corollary 1. *The vector $\boldsymbol{z}$ is the unique vector in $W^{\perp}$ that is "closest" to $\boldsymbol{y}$ i.e. $\operatorname{dist}(\boldsymbol{y}, W^{\perp}) = \|\boldsymbol{y} - \boldsymbol{z}\|_2$.*

Definition 5. *The $\boldsymbol{z}$ in Corollary 1 is called the Orthogonal Projection of $\boldsymbol{y}$ onto $W^{\perp}$.*

From now, we write $\hat{\mathbf{x}} :=$ the orthogonal projection of a vector $\mathbf{x} \in V$ onto $W^{\perp}$. Since $\hat{\mathbf{x}}$ is unique, so we can define a map $P : V \to W^{\perp}$ by $P(\mathbf{x}) = \hat{\mathbf{x}}$. It is easy to check that P is a linear map.

Theorem 5. *Let the columns of a $d \times k$ matrix $\boldsymbol{X}$ form a basis for a subspace $S^{\perp} \subseteq \mathbb{R}^d$. Then the matrix $\boldsymbol{P}$ is an orthogonal projection onto $S^{\perp}$ if and only if*

$$\boldsymbol{P} = \boldsymbol{X}\left(\boldsymbol{X}^{\top}\boldsymbol{X}\right)^{-1}\boldsymbol{X}^{\top}.$$

Proof. Let $\mathbf{v} \in \mathbb{R}^d$ and $\mathbf{P} = \mathbf{X}\left(\mathbf{X}^{\top}\mathbf{X}\right)^{-1}\mathbf{X}^{\top}$. Then $\hat{\mathbf{v}} = \mathbf{P}\mathbf{v} = \mathbf{X}\left(\mathbf{X}^{\top}\mathbf{X}\right)^{-1}\mathbf{X}^{\top}\mathbf{v} = \mathbf{X}\mathbf{u} \in S^{\perp}$ where $\mathbf{u} = \left(\mathbf{X}^{\top}\mathbf{X}\right)^{-1}\mathbf{X}^{\top}\mathbf{v}$. Now for any $\mathbf{y} \in \mathbb{R}^k$, we have $\mathbf{z} = \mathbf{X}\mathbf{y} \in S^{\perp}$ and by using $\langle \mathbf{z}, \mathbf{v} - \hat{\mathbf{v}} \rangle = \mathbf{z}^{\top}(\mathbf{v} - \hat{\mathbf{v}})$

$$\underbrace{(\mathbf{X}\mathbf{y})^{\top}}_{\mathbf{z}^{\top}} \underbrace{\left[\mathbf{v} - \mathbf{X}(\mathbf{X}^{\top}\mathbf{X})^{-1}\mathbf{X}^{\top}\mathbf{v}\right]}_{\mathbf{v}-\mathbf{P}\mathbf{v}} = \mathbf{y}^{\top}\left[\mathbf{X}^{\top}\mathbf{v} - \underbrace{\mathbf{X}^{\top}\mathbf{X}(\mathbf{X}^{\top}\mathbf{X})^{-1}}_{I_k}\mathbf{X}^{\top}\mathbf{v}\right] = 0.$$

Conversely, let $\mathbf{P}$ be an orthogonal projection onto S. We will prove that for any $\mathbf{v} \in \mathbb{R}^d$, $\hat{\mathbf{v}} = \mathbf{X}(\mathbf{X}^{\top}\mathbf{X})^{-1}\mathbf{X}^{\top}\mathbf{v}$. Note that $\mathbf{X}^{\top}\mathbf{X}$ is invertible. By definition $\hat{\mathbf{v}} \in S^{\perp} = \operatorname{Col}(\mathbf{X})$, so there is a vector $\mathbf{c} \in \mathbb{R}^d$ with $\mathbf{X}\mathbf{c} = \hat{\mathbf{v}}$. Also $\mathbf{v} - \hat{\mathbf{v}} = \mathbf{v} - \mathbf{X}\mathbf{c} \in S = \operatorname{Nul}(\mathbf{X}^{\top})$ as because $\operatorname{Nul}(\mathbf{X}^{\top})^{\perp} = \operatorname{Col}(\mathbf{X})$. So $\mathbf{0} = \mathbf{X}^{\top}(\mathbf{v} - \mathbf{X}\mathbf{c}) = \mathbf{X}^{\top}\mathbf{v} - \mathbf{X}^{\top}\mathbf{X}\mathbf{c} \implies \mathbf{c} = (\mathbf{X}^{\top}\mathbf{X})^{-1}\mathbf{X}^{\top}\mathbf{v}$. Hence $\hat{\mathbf{v}} = \mathbf{X}\mathbf{c} = \mathbf{X}(\mathbf{X}^{\top}\mathbf{X})^{-1}\mathbf{X}^{\top}\mathbf{v}$. $\square$

Corollary 2. *Let $\boldsymbol{U}$ be a $d \times k$ matrix with orthogonal columns and $S = \operatorname{span}(\boldsymbol{U})$. Then $\boldsymbol{P} = \boldsymbol{U}\boldsymbol{U}^{\top}$.*

Corollary 3. *$\boldsymbol{P}^2 = \boldsymbol{P}^{\top} = \boldsymbol{P}$. Moreover, $\boldsymbol{P}$ is similar to the diagonal matrix with k many 1s and $d - k$ many 0s.*

5.3.2 Projected Lattice.

Let $\mathcal{L} = \mathcal{L}(\mathbf{B})$,where $\mathbf{B} = \{\mathbf{b}_1, \mathbf{b}_2, \ldots, \mathbf{b}_k, \mathbf{b}_{k+1}, \ldots, \mathbf{b}_d\} \in \mathbb{R}^{d \times d}$ is invertible, be a lattice of rank d and $\tilde{\mathbf{B}} = \{\mathbf{b}_1^*, \mathbf{b}_2^*, \ldots, \mathbf{b}_k^*, \mathbf{b}_{k+1}^*, \ldots, \mathbf{b}_d^*\}$ be Gram-Schmidt orthogonalized basis. We define $k(< d)$-th Orthognal projection of a vector

$$\mathbf{v}(\in \mathbb{R}^d) = \sum_{i=1}^{d} c_i\, \mathbf{b}_i^*, c_i \in \mathbb{R}$$

with

$$\pi_k(\mathbf{v}) = \sum_{i=k}^{d} c_i\, \mathbf{b}_i^*$$

So it is easy to observe that we are projecting all the vectors $\mathbf{v}$ onto a $d - k + 1$ dimensional subspace having a basis $\{\mathbf{b}_k^*, \mathbf{b}_{k+1}^*, \ldots, \mathbf{b}_d^*\}$ and $\pi_k(\mathbf{b}_{k-1}) = \mathbf{0}$, $\pi_k(\mathbf{b}_k) = \mathbf{b}_k^*$, $\pi_k(\mathbf{b}_{k+1}) = \mathbf{b}_{k+1}^* + \mu_{k+1,k}\mathbf{b}_k^*$

and so on. Now we consider the set $\pi_k(\mathcal{L}) = \{\pi_k(\mathbf{v}) : \mathbf{v} \in \mathcal{L}\}$. Clearly the k-th projection of $\mathbf{0} \in \mathcal{L}$, is in $\pi_k(\mathcal{L})$. If $\mathbf{u}, \mathbf{v} \in \mathcal{L}$ then $c\pi_k(\mathbf{u}) - \pi_k(\mathbf{v}) = \pi_k(c\mathbf{u} - \mathbf{v}) \in \pi_k(\mathcal{L})$. Since $\mathcal{L}$ is discrete, so for any $\mathbf{u}, \mathbf{v} \in \mathcal{L}$,

$$\|\mathbf{u} - \mathbf{v}\|_2 = \ell(> 0) \implies \|\pi_k(\mathbf{u}) - \pi_k(\mathbf{v})\|_2 = \ell' > 0$$

So $\pi_k(\mathcal{L})$ is also a discrete subgroup of $\mathbb{R}^d$.

Definition 6. *We define π_k-th projection of a lattice $\mathcal{L}$ is $\pi_k(\mathcal{L}) = \{\pi_k(\boldsymbol{v}) : \boldsymbol{v} \in \mathcal{L}\}$.*

Lemma 3. *$\{\pi_k(\boldsymbol{b}_k), \pi_k(\boldsymbol{b}_{k+1}), \ldots, \pi_k(\boldsymbol{b}_d)\}$ is a basis of $\pi_k(\mathcal{L})$.*

Proof. Let $\mathcal{L} = \mathcal{L}(\mathbf{b}_1, \ldots, \mathbf{b}_d)$. Any $\mathbf{v} \in \mathcal{L}$ can be written as

$$\mathbf{v} = \sum_{i=1}^{d} c_i \mathbf{b}_i, \qquad c_i \in \mathbb{Z}.$$

Applying π_k gives

$$\pi_k(\mathbf{v}) = \sum_{i=k}^{d} c_i \pi_k(\mathbf{b}_i).$$

Since $\pi_k(\mathbf{b}_i) = \mathbf{0}$ for all $i < k$, we obtain

$$\pi_k(\mathbf{v}) = \sum_{i=k}^{d} c_i \pi_k(\mathbf{b}_i).$$

Thus every element of $\pi_k(\mathcal{L})$ is an integer linear combination of $\pi_k(\mathbf{b}_k), \ldots, \pi_k(\mathbf{b}_d)$, so they generate $\pi_k(\mathcal{L})$. Now let

$$\sum_{i=k}^{d} c_i \pi_k(\mathbf{b}_i) = \mathbf{0}$$

$$\implies c_k \mathbf{b}_k^* + c_{k+1}(\mathbf{b}_{k+1}^* + \mu_{k+1,k}\mathbf{b}_k^*) + \ldots + c_n(\mathbf{b}_n^* + \sum_{j<d}^{k} \mu_{d,j}\mathbf{b}_j^*) = \mathbf{0}$$

Assmbling all the coefficients of $\mathbf{b}_i^*$'s, and since $\{\mathbf{b}_k^*, \mathbf{b}_{k+1}^*, \ldots, \mathbf{b}_d^*\}$ is linearly independent, we have $c_n = c_{n-1} = \ldots = c_d = 0$. Hence $\{\pi_k(\mathbf{b}_k), \pi_k(\mathbf{b}_{k+1}), \ldots, \pi_k(\mathbf{b}_d)\}$ is a basis of $\pi_k(\mathcal{L})$. $\square$

Local projected block. In general, for integers i,j such that $1 \le i < j \le d$, the set $\mathbf{B}_{[i,j]} = \{\pi_i(\mathbf{b}_i), \pi_i(\mathbf{b}_{i+1}), \ldots, \pi_i(\mathbf{b}_j)\}$ is a basis of the projected lattice $\mathcal{L}_{[i,j]} := \pi_i(\mathcal{L}\{\mathbf{b}_i, \mathbf{b}_{i+1}, \ldots, \mathbf{b}_j\})$, where $\mathcal{L}(\mathbf{b}_i, \mathbf{b}_{i+1}, \ldots, \mathbf{b}_j)$ is a sub-lattice of $\mathcal{L}(\mathbf{B})$. Often, $\mathbf{B}_{[i,j]}$ is also called the *local projected block.*

5.4 BKZ

In this section, we see some basic notions and the workflow of the BKZ algorithm. The BKZ (Block Korkine–Zolotarev) algorithm is a lattice basis reduction algorithm that generalises LLL by operating on *blocks* of consecutive basis vectors rather than on pairs. Given a ordered basis $\mathbf{B} = (\mathbf{b}_1, \mathbf{b}_2, \ldots, \mathbf{b}_d)$ of a lattice $\mathcal{L} \subseteq \mathbb{R}^d$ and a *blocksize* $\beta \in [2, n]$, the goal of BKZ is to compute a β-BKZ-reduced basis, a notion that captures a strong quality guarantee on the Gram-Schmidt vectors of the output basis.

Definition 7. *Let $\boldsymbol{B} = (\boldsymbol{b}_1, \boldsymbol{b}_2, \ldots, \boldsymbol{b}_d)$ be an ordered basis and $\boldsymbol{B}^* = (\boldsymbol{b}_1^*, \boldsymbol{b}_2^*, \ldots, \boldsymbol{b}_d^*)$ be the corresponding Gram-Schmidt orthogonalized basis with Gram-Schmidt coefficient $\mu_{i,j} = \frac{\boldsymbol{b}_i \cdot \boldsymbol{b}_j^*}{\|\boldsymbol{b}_j^*\|^2}$, $1 \leq j < i \leq d$. The basis $\boldsymbol{B}$ is called size-reduced if:*

$$|\mu_{i,j}| \leq \frac{1}{2}.$$

Definition 8. *Let $2 \leq \beta \leq d$. A ordered basis $\boldsymbol{B} = (\boldsymbol{b}_1, \boldsymbol{b}_2, \ldots, \boldsymbol{b}_d)$ of a lattice $\mathcal{L}$ is said to be BKZ$-\beta$ reduced if $\boldsymbol{B}$ is size-reduced and $\|\boldsymbol{b}_i^*\|_2 = \lambda_1\left(\mathcal{L}_{[i,\min(i+\beta-1,d)]}\right)$ for each $1 \leq i \leq d-1$. The number β is called the blocksize. In particular,*

$$\|\boldsymbol{b}_i^*\|_2 = \begin{cases} \lambda_1\big(\mathcal{L}_{[i,i+\beta-1]}\big)\,, & \text{if } 1 \leq i \leq d-\beta+1, \\ \lambda_1\big(\mathcal{L}_{[i,d]}\big)\,, & \text{if } d-\beta+2 \leq i \leq d-1. \end{cases}$$

The BKZ algorithm (Algorithm 1) maintains two data structures throughout: the basis $\mathbf{B} = (\mathbf{b}_1, \ldots, \mathbf{b}_d)$ and the Gram–Schmidt triangular matrix $\mathbf{U}$ (containing the $\mu_{i,j}$ coefficients and norms $\|\mathbf{b}_i^*\|_2^2$). It terminates by detecting when a full round produces no changes happen in the basis $\mathbf{B}$.

Algorithm 1: Block Korkine–Zolotarev (BKZ) — [Che13, Algorithm 18]

Input: Basis $\mathbf{B} = (\mathbf{b}_1, \ldots, \mathbf{b}_d)$; targeted β
Output: Basis $(\mathbf{b}_1, \ldots, \mathbf{b}_d)$ is BKZ-β reduced
1 $(\mathbf{B}, \mathbf{U}) \leftarrow \mathrm{LLL}(\mathbf{b}_1, \ldots, \mathbf{b}_d, \mathbf{U})$; `// LLL-reduce and compute Gram--Schmidt matrix U`
2 **while** *$\boldsymbol{B}$ is not fully changed* **do**
3 $\quad (\mathbf{B}, \mathbf{U}) \leftarrow \textsc{OneRoundBKZ}(\beta, \mathbf{B}, \mathbf{U})$;
4 return $\mathbf{B}$

One round of BKZ sweeps the sliding window $j = 1, 2, \ldots, d-1$ once. At each position j, it defines the local block $\mathbf{B}_{[j,k]}$ with $k = \min(j+\beta-1,\, d)$, calls an enumeration oracle to find the shortest vector in $\mathcal{L}_{[j,k]}$, and either inserts a new (shorter) basis vector or simply LLL-prepares the next block.

The high-level algorithm. The BKZ algorithm begins by computing an LLL-reduced basis $\mathbf{B} \leftarrow \mathrm{LLL}(\mathbf{B})$. It then *repeatedly* performs *tours* or *rounds* through the basis until the basis no longer changes. Algorithmically, we call it OneRoundBKZ and works as a subroutine of BKZ. There are $d-1$ projected blocks. The OneRoundBKZ acts on first $d-\beta+1$ projected blocks $\mathbf{B}_{[1,\beta]}$, $\mathbf{B}_{[2,\beta+1]}$, $\ldots$, $\mathbf{B}_{[d-\beta+1,d]}$ of length β and $\beta-2$ projected blocks $\mathbf{B}_{[d-\beta+2,d]}$, $\mathbf{B}_{[d-\beta+3,n]}$, $\ldots$, $\mathbf{B}_{[d-1,d]}$, whose lengths decrease from $\beta-1$ down to 2. The BKZ algorithm terminates and returns $\mathbf{B}$ once a full tour produces no change to the basis.

OneRoundBKZ. The core building block of OneRoundBKZ is the operations on each $\mathbf{B}_{[i,j]}$. Like the LLL swap operation, the OneRoundBKZ modifies the lattice basis locally to prevent the Gram-Schmidt vectors from shrinking too rapidly; however, while the LLL swap acts on two adjacent basis vectors, the OneRoundBKZ works on a block of up to β consecutive basis vectors. The workflow of the algorithm proceeds as follows:

1. **Find a shortest vector.** An SVP oracle (either a state-of-the-art enumeration algorithm or a lattice sieving algorithm) is called on the projected lattice $\mathcal{L}_{[j,k]}$ to find a shortest nonzero vector $\mathbf{v}' \in \mathcal{L}_{[j,k]}$.

Algorithm 2: OneRoundBKZ — [Che13, Algorithm 19]

Input: Basis $\mathbf{B} = (\mathbf{b}_1, \ldots, \mathbf{b}_d)$; norms $\|\mathbf{b}_1^*\|_2^2, \ldots, \|\mathbf{b}_d^*\|_2^2$; blocksize $\beta \in \{2, \ldots, d\}$; inputs related to `SVP-ORACLE`

Output: Updated basis $(\mathbf{b}_1, \ldots, \mathbf{b}_d)$ after one BKZ-β round; flag `f` indicating whether `SVP-ORACLE` succeeded

$j \leftarrow 0$;

`f` $\leftarrow$ UNSUCCESSFUL;

while $j < d - 1$ **do**

  $j \leftarrow (j \bmod (d-1)) + 1$; $k \leftarrow \min(j + \beta - 1,\, d)$; $h \leftarrow \min(k+1,\, d)$; `// Define the local block` $\mathcal{L}_{[j,k]}$

  $\mathbf{v} \leftarrow$ `SVP-ORACLE`$\big(\mathbf{B}_{[j,k]}\big)$; `// Find` $\mathbf{v} = (v_j, \ldots, v_k) \in \mathbb{Z}^{k-j+1} \setminus \{\mathbf{0}\}$ `s.t.` $\|\pi_j(\sum_{i=j}^{k} v_i \mathbf{b}_i)\|_2 = \lambda_1(\mathcal{L}_{[j,k]})$

  if $\|\boldsymbol{v}^*\|_2 < \|\boldsymbol{b}_j^*\|_2$ **then**

    `f` $\leftarrow$ SUCCESSFUL;

    LLL$\Big(\mathbf{b}_1, \ldots, \mathbf{b}_{j-1}, \sum_{i=j}^{k} v_i \mathbf{b}_i,\ \mathbf{b}_j, \ldots, \mathbf{b}_h,\ \mathbf{U}\Big)$ at stage j; `// Insert new vector at start of block; remove dependency; update` μ

  else

    LLL$(\mathbf{b}_1, \ldots, \mathbf{b}_h,\ \mathbf{U})$ at stage $h-1$;

return $\mathbf{B}$

2. **Lift to a lattice vector.** Writing $\mathbf{v}' = \sum_{i=j}^{k} v_i \pi_j(\mathbf{b}_i)$ with $v_i \in \mathbb{Z}$, the corresponding full lattice vector is

$$\mathbf{v} \;=\; \sum_{i=j}^{k} v_i \mathbf{b}_i \;\in\; \mathcal{L}.$$

3. **Set the next block index.** Set $h \leftarrow \min(k+1, d)$, so that the subsequent BKZ block operation will act on $\mathbf{B}_{[j+1,h]}$.

4. **Decide whether to insert v.** Compute the Gram-Schmidt vector $\mathbf{v}^* = \mathbf{v} - \sum_{\ell<j} \frac{\mathbf{v}\cdot\mathbf{b}_\ell^*}{\|\mathbf{b}_\ell^*\|_2^2} \mathbf{b}_\ell^*$ corresponding to position j in the extended set $(\mathbf{b}_1, \ldots, \mathbf{b}_{j-1}, \mathbf{v}, \mathbf{b}_j, \ldots, \mathbf{b}_h)$. By putting $\mathbf{b}_i = \mathbf{b}_i^* + \sum_{\ell<i} \mu_{i,\ell} \mathbf{b}_l^*$ in the expression of $\mathbf{v}$ and $\mathbf{v}^*$, we have

$$\begin{aligned}\mathbf{v}^* &= (v_j + v_{j+1}\mu_{j+1,j} + \cdots + v_k \mu_{k,j})\mathbf{b}_j^* \\ &\quad + (v_{j+1} + v_{j+2}\mu_{j+2,j+1} + \cdots + v_k \mu_{k,j+1})\mathbf{b}_{j+1}^* \\ &\quad + \cdots + v_k \mathbf{b}_k^* \\ &= \pi_j(\mathbf{v}) \\ &= \mathbf{v}'\end{aligned}$$

Since $\mathbf{v}'$ is one of the shortest vectors in $\mathcal{L}_{[j,k]}$ and $\pi_j(\mathbf{b}_j) = \mathbf{b}_j^* \in \mathcal{L}_{[j,k]}$, hence

$$\|\mathbf{v}^*\|_2 \;=\; \|\pi_j(\mathbf{v})\|_2 \;\le\; \|\pi_j(\mathbf{b}_j)\|_2 \;=\; \|\mathbf{b}_j^*\|_2, \tag{8}$$

the j-th Gram-Schmidt vector of this extended set is at most as long as $\mathbf{b}_j^*$, and the first $j-1$ Gram-Schmidt vectors remain unchanged. If $\|\mathbf{v}^*\|_2 < \|\mathbf{b}_j^*\|_2$, the vector $\mathbf{v}$ is strictly shorter and is worth inserting. In that case, $\mathbf{v}$ is placed between $\mathbf{b}_{j-1}$ and $\mathbf{b}_j$, producing the ordered (but linearly *dependent*) set

$$\mathbf{B}' = (\mathbf{b}_1, \ldots, \mathbf{b}_{j-1}, \mathbf{v}, \mathbf{b}_j, \ldots, \mathbf{b}_h).$$

Linear independence of $\mathbf{B}'$ is then restored by applying a *modified-LLL* procedure to the dependent vectors $(\mathbf{b}_1, \ldots, \mathbf{b}_{j-1}, \mathbf{v}, \mathbf{b}_j, \ldots, \mathbf{b}_h)$, yielding a linearly independent (and LLL-reduced) sub-basis $\hat{\mathbf{B}}$:

$$\hat{\mathbf{B}} \leftarrow \text{modified-LLL}(\mathbf{b}_1, \ldots, \mathbf{b}_{j-1}, \mathbf{v}, \mathbf{b}_j, \ldots, \mathbf{b}_h).$$

On the other hand, if $\|\mathbf{v}^*\|_2 = \|\mathbf{b}_i^*\|_2$, the block $(\mathbf{b}_1, \ldots, \mathbf{b}_h)$ is already LLL-reduced at this position, and a standard LLL call suffices:

$$\hat{\mathbf{B}} \leftarrow \text{LLL}(\mathbf{b}_1, \ldots, \mathbf{b}_h).$$

5. **Reconstruct the full basis.** The updated full basis is assembled as

$$\mathbf{B} \leftarrow \hat{\mathbf{B}} \cup (\mathbf{b}_{h+1}, \ldots, \mathbf{b}_d),$$

where the tail $(\mathbf{b}_{h+1}, \ldots, \mathbf{b}_d)$ is left unchanged, and the function returns $\mathbf{B}$.

5.5 A new estimate and Primal Attack

Formally, we define Primal Attack as defined in [ADPS16], which consists of constructing a unique-SVP instance from the Short-Secret LWE problem and solving it using BKZ. Given an Short-Secret LWE instance $(\mathbf{b}, \mathbf{b} = \mathbf{A}^\top \mathbf{s} + \mathbf{e})$, one can construct a lattice

$$\mathcal{L} = \{\mathbf{x} \in \mathbb{Z}^{m+n+1} : (\mathbf{I}_m | \mathbf{A}^\top_{m \times n} | -\mathbf{b})\mathbf{x} \equiv \mathbf{0} \mod q\}$$

consisting a basis $\mathbf{B} = \begin{bmatrix} q\mathbf{I}_m & -\mathbf{A}^\top & \mathbf{b} \\ 0 & \mathbf{I}_n & 0 \\ 0 & 0 & 1 \end{bmatrix}$. Note that $\mathrm{Vol}(\mathcal{L}) = q^m$ and Λ contains a unique SVP solution $\begin{bmatrix}\mathbf{e} \mid \mathbf{s} \mid 1\end{bmatrix}^\top$. The BKZ-$\beta$ will find $\begin{bmatrix}\mathbf{e} \mid \mathbf{s} \mid 1\end{bmatrix}^\top$ if and only if

$$|| \begin{bmatrix}\mathbf{e} \mid \mathbf{s} \mid 1\end{bmatrix}^\top ||_2 \sqrt{\frac{\beta}{d}} \approx \sigma\sqrt{\beta} \leq \delta^{2\beta-d}\,\mathrm{Vol}(\Lambda)^{\frac{1}{d}} \tag{9}$$

where $d = m+n+1$ is the dimension of the lattice Λ and β is block size. The success condition of BKZ is called "**Estimate 2016**". This estimate had been investigated extensively by Albrecht et al. in [AGVW17]. They also show that the lattice reduction experiments largely follow the behaviour expected from the "**Estimate 2016**".

5.6 Why the Estimate 2008 is Wrong

The earlier 2008 condition (Gama–Nguyen [GN08]) used:

$$\frac{\lambda_2(\mathcal{L})}{\lambda_1(\mathcal{L})} \geq c \cdot \delta^d,$$

asking whether the *global gap* between the first and second minimum of the *whole* lattice is large enough.

The flaw is that BKZ does not operate on the whole lattice at once. It calls the SVP oracle only on a β-dimensional *projected block*. The correct question is therefore not

"Is $\mathbf{v}$ short relative to the whole lattice?"

but rather

"Is the *projection* of $\mathbf{v}$ short relative to just that β-dimensional block?"

The 2016 condition (9) asks precisely the latter. Since the projected lattice $\mathcal{L}_{[d-\beta+1,d]}$ has a much smaller volume than $\mathcal{L}$ itself, the bar for detection is significantly lower, explaining why the "**Estimate 2016**" predicts cheaper attacks than the 2008 one.

5.7 Justification of condition (9)

Definition 9 (Geometric Series Assumption). *The norms of the Gram-Schmidt vectors after lattice reduction satisfy*

$$\|\boldsymbol{b}_i^*\|_2 = \alpha^{i-1} \cdot \|\boldsymbol{b}_1\|_2 \quad \textit{for some } 0 < \alpha < 1 \tag{10}$$

By Equation 4 we have $\|\mathbf{b}_1\|_2 = \delta^d \cdot \mathrm{Vol}(\mathcal{L})^{\frac{1}{d}}$ and $\mathrm{Vol}(\mathcal{L}) = \prod_{i=1}^{d} \|\mathbf{b}_i^*\|_2$ as we know from the Gram-Schmidt Orthogonalization $\mathbf{B} = \mathbf{L} \cdot \tilde{\mathbf{B}}$, we get

$$\begin{aligned}
\mathrm{Vol}(\mathcal{L}) &= \|\mathbf{b}_1^*\|_2 \cdot \|\mathbf{b}_2^*\|_2 \ldots \cdot \|\mathbf{b}_d^*\|_2 &(11)\\
&= \|\mathbf{b}_1^*\|_2 \cdot \alpha \|\mathbf{b}_1^*\|_2 \ldots \cdot \alpha^{d-1} \|\mathbf{b}_1^*\|_2 &(12)\\
&= \alpha^{1+2+\ldots+(d-1)} \cdot \|\mathbf{b}_1^*\|_2^d &(13)\\
&= \alpha^{d(d-1)/2} \cdot \delta^{d^2} \cdot \mathrm{Vol}(\mathcal{L}) &(14)\\
\therefore \alpha &= \delta^{\frac{-2d}{d-1}} \approx \delta^{-2} \text{ for the GSA.} &(15)
\end{aligned}$$

In [ADPS16], the considered LWE has a short secret, the secret is drawn from the error distribution χ centred around 0, and uses the new attack by Bai-Galbraith. The uSVP solution will be an embedded vector for which each entry is drawn i.i.d. from a distribution of standard deviation σ and mean $\mu = 0$ with additional constant entry 1, and the solution is $\mathbf{v} = \left[\mathbf{e} \mid \mathbf{s} \mid 1\right]^\top$. Because each component of the secret and the error vectors comes from the distribution χ, we do not need to consider rebalancing. Suppose the distribution χ is the uniform distribution on $[\![-a, a]\!]$. So χ has mean $c = 0$ and variance:

$$\frac{1}{2a+1} \sum_{i=-a}^{a} i^2 = \frac{a(a+1)}{3} \implies \sigma = \sqrt{\frac{a(a+1)}{3}}$$

Suppose each component of the error and the secret vector are coming from $D_{[\![-a,a]\!],\sigma=\sqrt{\frac{a(a+1)}{3}},c=0}$. Then $\|\mathbf{v}\|_2^2$ follows a scaled chi-squared distribution $\sigma^2 \cdot \chi_{d-1}^2$ with $d-1$ degress of freedom, with a fixed 1, resulting in $\mathrm{E}[\|\mathbf{v}\|_2^2] = (d-1)\sigma^2 + 1 \approx d \cdot \sigma^2$. By Theorem 5, any k-th projection has the form $\mathbf{P} = \mathbf{X}\big(\mathbf{X}^\top\mathbf{X}\big)^{-1}\mathbf{X}^\top$, where the columns of $\mathbf{X}$ form a k dimensional subspace. The $d \times k$ matrix $\mathbf{X}$ has singular value decomposition

$$\mathbf{X} = \mathbf{UDV}$$

where $\mathbf{U}$ is a $d \times d$ orthonormal, $\mathbf{D}$ is a $d \times k$ diagonal, and $\mathbf{V}$ is a $k \times k$ orthonormal matrix. So the matrix product

$$\mathbf{X}^\top\mathbf{X} = \mathbf{V}^\top\mathbf{D}^2\mathbf{V},$$

whose inverse is

$$(\mathbf{X}^\top \mathbf{X})^{-1} = \mathbf{V}^\top \mathbf{D}^{-2} \mathbf{V},$$

if $\mathbf{D}$ is non-singular, that is, if $\mathbf{X}$ is basis.

The matrix $\mathbf{P}$ is then

$$\mathbf{X}^\top \mathbf{V}^\top \mathbf{D}^{-2} \mathbf{V} \mathbf{X} = \mathbf{U}\mathbf{U}^\top.$$

The distribution of $\mathbf{Pv}$ depends only on the length of any vector $\mathbf{v}$ because for each fixed $k \times k$ orthogonal matrix $\mathbf{B}$, the matrix $\mathbf{UB}$ is distributed the same way $\mathbf{U}$ is. It is a consequence of the rotational symmetry built into the $\mathcal{N}(\mathbf{0}, \mathbf{I}_{k\times k})$ distribution. Thus,

$$\mathbb{E}[\|\mathbf{Pv}\|_2^2] = \|\mathbf{v}\|_2^2 \, \mathbb{E}[\mathrm{P}_{11}],$$

as we can take

$$\mathbf{v} = (\|\mathbf{v}\|_2, 0, 0, \ldots, 0)^\top.$$

But what is $\mathbb{E}[\mathrm{P}_{11}]$? Since the distribution of $\mathbf{U}$ is rotation invariant in $\mathbb{R}^d$, so it means the distributions of all diagonal entries $\mathrm{P}_{ii}, 1 \leq i \leq d$ are identical, so the expectations are equal. By using Lemma 3

$$\mathrm{tr}(\mathbf{P}) = \mathrm{tr}(\mathbf{U}\mathbf{U}^\top) = k.$$

Finally

$$\mathbb{E}[\mathrm{P}_{11}] = \frac{k}{d}, \quad \mathbb{E}[\|\mathbf{Pv}\|_2^2] = \|\mathbf{v}\|_2^2 \, \mathbb{E}[\mathrm{P}_{11}] = \|\mathbf{v}\|_2^2 \frac{k}{d}.$$

Expected norm of the projection of v. As $\pi_{d-\beta+1}(\mathbf{v}) \in \mathcal{L}_{[d-\beta+1,d]}$ and $\mathcal{L}_{[d-\beta+1,d]}$ is $\beta-$dimensional projected lattice, the expected ℓ_2 norm of $\pi_{d-\beta+1}(\mathbf{v}) = \frac{\sqrt{\beta}}{\sqrt{d}} \|\mathbf{v}\|_2$. But we have assumed all the components of $\mathbf{v} = \begin{bmatrix} \mathbf{e} \mid \mathbf{s} \mid 1 \end{bmatrix}^\top$ is coming from $D_{[\![-a,a]\!],\sigma=\sqrt{\frac{a(a+1)}{3}},c=0}$, so the expected ℓ_2 norm of $\mathbf{v}$ is $\sigma\sqrt{d}$ and hence $\mathbb{E}[\|\pi_{d-\beta+1}(\mathbf{v})\|_2] = \sigma\sqrt{\beta}$.

GSA prediction for the last block. After BKZ-β reduction, the Gram-Schmidt vectors obey that GSA assumption, then

$$\|\mathbf{b}^*_{d-\beta+1}\|_2 = \alpha^{d-\beta+1-1} \|\mathbf{b}_1\|_2 = \delta^{2\beta-d} \cdot \mathrm{Vol}(\mathcal{L})^{\frac{1}{d}}$$

by using (4) and (15). By the Gaussian Heuristic, the shortest vector in the projected lattice $\mathcal{L}_{[d-\beta+1,d]}$ is approximately $\|\mathbf{b}^*_{d-\beta+1}\|_2$, so:

$$\lambda_1(\mathcal{L}_{[d-\beta+1,d]}) \approx \delta^{2\beta-d} \cdot \mathrm{Vol}(\mathcal{L})^{\frac{1}{d}}.$$

The SVP oracle, when called on the last block, finds the shortest vector in $\mathcal{L}_{[d-\beta+1,d]}$. It will find $\pi_{d-\beta+1}(\mathbf{v})$ and BKZ inserts it in between $\mathbf{b}^*_{d-\beta}$ and $\mathbf{b}^*_{d-\beta+1}$. By (8), it is now clear that

$$\|\pi_{d-\beta+1}(\mathbf{v})\|_2 \leq \|\mathbf{b}^*_{d-\beta+1}\|_2 \implies \sigma\sqrt{\beta} \leq \delta^{2\beta-d} \cdot \mathrm{Vol}(\mathcal{L})^{\frac{1}{d}}.$$

and we derived condition (9). It remains to be seen whether condition (9) holds and how the entire unique shortest vector can be recovered. We will discuss all the steps throughout the next sections.

5.8 Finding the Projection

5.8.1 The projection lives in the sub-lattice

Since $\mathbf{v} = \sum_{i=1}^{d} v_i \mathbf{b}_i$ with $v_i \in \mathbb{Z}$, we have

$$\pi_{d-\beta+1}(\mathbf{v}) = \sum_{i=d-\beta+1}^{d} v_i \, \pi_{d-\beta+1}(\mathbf{b}_i) \ \in \ \mathcal{L}_{[d-\beta+1,d]}.$$

So the projection is a genuine lattice vector in the last projected block.

5.8.2 Why the SVP oracle finds it

When condition (9) holds:

$$\sigma\sqrt{\beta} \leq \delta^{2\beta-d} \cdot \mathrm{Vol}(\mathcal{L})^{\frac{1}{d}}$$

so $\pi_{d-\beta+1}(\mathbf{v})$ is at most as long as the expected shortest vector in that sub-block under the GSA. The SVP oracle, therefore, finds $\pm\pi_{d-\beta+1}(\mathbf{v})$ and BKZ inserts a new basis vector

$$\mathbf{b}_{d-\beta+1}^{\mathrm{new}} = \pm \sum_{i=d-\beta+1}^{d} v_i \, \mathbf{b}_i$$

at position $d - \beta + 1$, recovering the last β coefficients $v_{d-\beta+1}, \ldots, v_d$ of $\mathbf{v}$.

5.8.3 Propagation across BKZ tours

The argument extends recursively across tours. Once $\pi_{d-\beta+1}(\mathbf{v})$ is found in tour 1, tour 2 can find $\pi_{d-2\beta+2}(\mathbf{v})$ using the updated basis, and so on:

Tour 1:	$\pi_{d-\beta+1}(\mathbf{v})$ found	(coefficients $v_{d-\beta+1}, \ldots, v_d$)
Tour 2:	$\pi_{d-2\beta+2}(\mathbf{v})$ found	(coefficients $v_{d-2\beta+2}, \ldots, v_{d-\beta}$)
	$\vdots$	

In practice, however, *size reduction completes the full recovery in the same step as the projection is found* (Section 5.9).

5.9 Stage 2: Size Reduction Recovers v

5.9.1 Setup after finding the projection

After BKZ inserts $\mathbf{b}_{d-\beta+1}^{\mathrm{new}}$, the basis satisfies $\mathbf{b}_{d-\beta+1}^{*} = \pi_{d-\beta+1}(\mathbf{v})$, and $\mathbf{v}$ decomposes as

$$\mathbf{v} = \mathbf{b}_{d-\beta+1}^{\mathrm{new}} + \sum_{i=1}^{d-\beta} v_i \, \mathbf{b}_i.$$

The remaining unknowns are the coefficients $v_1, \ldots, v_{d-\beta}$ corresponding to the "front" of the basis.

5.9.2 The size reduction iteration

Size reduction walks *backwards* from $i = d-\beta$ down to $i = 1$. Maintain a running vector initialised as

$$\mathbf{b}_{d-\beta+1}^{(d-\beta+1)} := \mathbf{b}_{d-\beta+1}^{\text{new}}.$$

At each step i, define the integer coset

$$L_i := \left\{\, \mu\, \mathbf{b}_i^* + \pi_i\!\left(\mathbf{b}_{d-\beta+1}^{(i+1)}\right) \;\middle|\; \mu \in \mathbb{Z} \,\right\},$$

and find the integer $\mu_i \in \mathbb{Z}$ minimising the element of L_i, i.e. the element closest to the origin. Set

$$\mathbf{b}_{d-\beta+1}^{(i)} := \mu_i\, \mathbf{b}_i + \mathbf{b}_{d-\beta+1}^{(i+1)}.$$

5.9.3 The coset contains $\pi_i(\mathbf{v})$

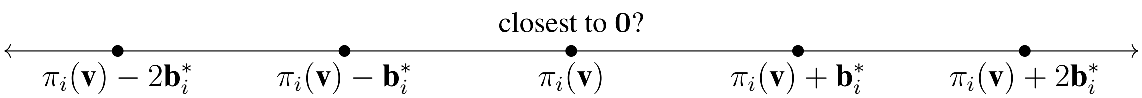
An important observation is that $\pi_i(\mathbf{v})$ belongs to L_i:

$$\pi_i(\mathbf{v}) = v_i\, \mathbf{b}_i^* + \pi_i\!\left(\mathbf{b}_{d-\beta+1}^{(i+1)}\right) \;\in\; L_i,$$

so we can rewrite the coset as

$$L_i = \pi_i(\mathbf{v}) + \mathbb{Z}\, \mathbf{b}_i^*.$$

This is a one-dimensional integer coset, i.e. equally spaced points along direction $\mathbf{b}_i^*$ centred at $\pi_i(\mathbf{v})$:



Size reduction finds $\mu_i = v_i$ (and hence sets $\mathbf{b}_{d-\beta+1}^{(i)} = \mathbf{b}_{d-\beta+1}^{(i+1)} + v_i\, \mathbf{b}_i$) **if and only if** $\pi_i(\mathbf{v})$ is the closest point in L_i to the origin:

$$\|\pi_i(\mathbf{v})\|_2 < \min\{\|\pi_i(\mathbf{v}) + \mathbf{b}_i^*\|_2,\ \|\pi_i(\mathbf{v}) - \mathbf{b}_i^*\|_2\}\,. \tag{16}$$

If (16) holds at every step $i = d-\beta, \ldots, 1$, then

$$\mathbf{b}_{d-\beta+1}^{(1)} = \mathbf{b}_{d-\beta+1}^{\text{new}} + \sum_{i=1}^{d-\beta} v_i\, \mathbf{b}_i = \mathbf{v},$$

so size reduction has recovered $\mathbf{v}$ exactly.

5.10 Why the Success Condition Guarantees Size Reduction Works

5.10.1 Probability of success at step i

At step i, let $r_i := \|\pi_i(\mathbf{v})\|_2$ and $R_i := \|\mathbf{b}_i^*\|_2$. Under the assumption that $\pi_i(\mathbf{v})$ points in a uniformly random direction (which holds for LWE where $(\mathbf{e}, \mathbf{s})$ is drawn from a near-spherical distribution), the probability that (16) holds is

$$p_i \;\approx\; \begin{cases} 1 - \dfrac{2\, A_{d-i+1}(r_i,\, h_i)}{A_{d-i+1}(r_i)} & \text{if } R_i < 2r_i, \\ 1 & \text{if } R_i \geq 2r_i, \end{cases} \tag{17}$$

where

- $A_{d-i+1}(r_i)$ is the surface area of the sphere in $\mathbb{R}^{d-i+1}$ with radius r_i,
- $A_{d-i+1}(r_i, h_i)$ is the area of the hyperspherical cap of height $h_i = r_i - R_i/2$,
- the condition $R_i \geq 2r_i$ corresponds exactly to the classical size reduction guarantee (all coordinates of **v** in the current basis are at most $1/2$ in absolute value).

Using the formula from [Li11], this expands to

$$p_i \approx \begin{cases} 1 - \dfrac{\displaystyle\int_0^{2\frac{h_i}{r_i} - \left(\frac{h_i}{r_i}\right)^2} t^{(d-i)/2-1}(1-t)^{-1/2}\, dt}{B\left(\frac{d-i}{2}, \frac{1}{2}\right)} & \text{if } R_i < 2r_i, \\ 1 & \text{if } R_i \geq 2r_i, \end{cases}$$

where $B(\cdot, \cdot)$ denotes the Euler beta function.

5.10.2 Overall success probability

The total probability that size reduction recovers **v** from $\pi_{d-\beta+1}(\mathbf{v})$ is:

$$p = \prod_{i=1}^{d-\beta} p_i. \tag{18}$$

This product is close to 1 for all $\beta \gtrsim 30$, because:

- For small i (early basis vectors), $\|\mathbf{b}_i^*\|_2$ is large relative to $\|\pi_i(\mathbf{v})\|_2$, so $R_i \geq 2r_i$ and $p_i = 1$ exactly.
- For i near $d-\beta$, condition (9) controls the ratio r_i/R_i, keeping p_i close to 1.

Furthermore, when condition (9) holds with equality, the ratio r_i/R_i at each step depends only on β (not on d, σ, or $\mathrm{Vol}(\mathcal{L})$ individually), so p is a function of β alone.

5.11 The Formal Claim

Claim 1 (Albrecht et al.). *Let $\boldsymbol{v} \in \mathcal{L} \subset \mathbb{R}^d$ be a unique shortest vector and $\beta \in \mathbb{N}$. Assume that* (9) *holds, the current basis $\boldsymbol{B} = \{\boldsymbol{b}_1, \ldots, \boldsymbol{b}_d\}$ satisfies $\boldsymbol{b}^*_{d-\beta+1} = \pi_{d-\beta+1}(\boldsymbol{v})$, and*

$$\boldsymbol{v} = \boldsymbol{b}^{\mathrm{new}}_{d-\beta+1} + \sum_{i=1}^{d-\beta} v_i\, \boldsymbol{b}_i, \quad v_i \in \mathbb{Z},$$

and the GSA holds for $\boldsymbol{B}$ up to index $d-\beta+1$. If the size reduction step of BKZ-β is called on $\boldsymbol{b}^{\mathrm{new}}_{d-\beta+1}$, it recovers $\boldsymbol{v}$ with high probability over the randomness of the basis.

Phase	What happens
GSA established	BKZ reduces the basis until Gram–Schmidt norms follow the geometric profile $\|\mathbf{b}_i^*\|_2 = \delta^{d-2i+1} \cdot \mathrm{Vol}(\mathcal{L})^{\frac{1}{d}}$.
SVP oracle call	Called on the last β-dimensional projected block. Since $\|\pi_{d-\beta+1}(\mathbf{v})\|_2 \leq \delta^{2\beta-d} \cdot \mathrm{Vol}(\mathcal{L})^{\frac{1}{d}}$ by (9), the projection stands out as anomalously short and is returned by the oracle.
Basis update	BKZ inserts $\mathbf{b}_{d-\beta+1}^{\text{new}} = \pm \sum_{i=d-\beta+1}^{d} v_i \mathbf{b}_i$ into position $d-\beta+1$, recovering the last β coefficients of $\mathbf{v}$.
Size reduction	Walking from $i = d-\beta$ down to 1, size reduction finds $\mu_i = v_i$ at each step by minimising in $L_i = \pi_i(\mathbf{v}) + \mathbb{Z}\mathbf{b}_i^*$, yielding $\mathbf{b}_{d-\beta+1}^{(1)} = \mathbf{v}$ with probability $p \approx 1$.
Output	$\mathbf{v} = [\mathbf{e} \mid \mathbf{s} \mid 1]^\top$ is in the basis; the LWE secret is recovered.

Table 3: End-to-end BKZ recovery of the unique shortest vector.

5.12 Summary

Key insight in one sentence. Condition (9) ensures the projection of **v** into the last BKZ block is anomalously short (so the SVP oracle finds it), and from that projection, size reduction deterministically unwinds the remaining coefficients one by one because the GSA guarantees that earlier basis vectors are long enough to make $\pi_i(\mathbf{v})$ always the closest coset element to the origin.

To estimate the cost of the attack, one has to find an optimal BKZ block size β and an optimal $\mathbf{LWE}_{m,n,q,a}$ sample size m so that the unique shortest vector can be recovered from the reduced lattice basis under the GSA. The reader can visit this link to find an $\mathbf{LWE}_{m,n,q,a}$ estimator used to estimate parameters for the primal attack on CRYSTALS-Kyber and CRYSTALS-Dilithium.

Note. *Suppose we have a matrix $\boldsymbol{A} \in \mathbb{Z}_q^{n\times m}, n \leq m$ with rank$(\boldsymbol{A}) = n$. Consider the q-ary lattice $\Lambda_q^{\perp}(\boldsymbol{A}) = \{\boldsymbol{x} \in \mathbb{Z}^m : \boldsymbol{A}\boldsymbol{x} \equiv 0 \mod q\}$. $\boldsymbol{A} = [\boldsymbol{A}_1|\boldsymbol{A}_2]$ where $\boldsymbol{A}_1$ is invertible of order $n\times n$ and $\boldsymbol{A}_2$ is of order $n\times(m-n)$. We are interested in finding a basis for this lattice. Let $\boldsymbol{x} \in \Lambda_q^{\perp}(\boldsymbol{A})$.*

$$\boldsymbol{A}\boldsymbol{x} = [\boldsymbol{A}_1|\boldsymbol{A}_2]\boldsymbol{x} \equiv \boldsymbol{0} \mod q$$

then,

$$\boldsymbol{A}_1^{-1}\boldsymbol{A}\boldsymbol{x} = [\boldsymbol{I}|\boldsymbol{A}_1^{-1}\boldsymbol{A}_2]\boldsymbol{x} = q\boldsymbol{u}, \boldsymbol{u} \in \mathbb{Z}^n$$

implies,

$$\boldsymbol{x}_1 + \boldsymbol{A}_1^{-1}\boldsymbol{A}_2\boldsymbol{x}_2 = q\boldsymbol{u}, \textit{where } \boldsymbol{x} = \begin{bmatrix}\boldsymbol{x}_1 \mid \boldsymbol{x}_2\end{bmatrix}^{\top}$$

So,

$$\boldsymbol{x} = \begin{bmatrix}\boldsymbol{x}_1\\ \boldsymbol{x}_2\end{bmatrix} = \begin{bmatrix}q\boldsymbol{u} - \boldsymbol{A}_1^{-1}\boldsymbol{A}_2\boldsymbol{x}_2\\ \boldsymbol{x}_2\end{bmatrix} = \begin{bmatrix}q\boldsymbol{I}_n & -\boldsymbol{A}_1^{-1}\boldsymbol{A}_2\\ \boldsymbol{0} & \boldsymbol{I}_{m-n}\end{bmatrix}\begin{bmatrix}\boldsymbol{u}\\ \boldsymbol{x}_2\end{bmatrix}$$

$\boldsymbol{B}^{'} = \begin{bmatrix}q\boldsymbol{I}_n & -\boldsymbol{A}_1^{-1}\boldsymbol{A}_2\\ \boldsymbol{0} & \boldsymbol{I}_{m-n}\end{bmatrix}$ *is a basis and volume of the lattice $\Lambda_q^{\perp}(\boldsymbol{A})$ is q^n. The* GSO $\left(\boldsymbol{B}^{'}\right)$ *consists of*

- *For $i = 1, \ldots, n$:*

$$\boldsymbol{v}_i^{*} = \begin{bmatrix}q\boldsymbol{e}_i\\ \boldsymbol{0}\end{bmatrix}, \qquad \|\boldsymbol{v}_i^{*}\|_2 = q.$$

- *For $t = 1, \ldots, m-n$:*

$$\boldsymbol{v}_{n+t}^{*} = \begin{bmatrix}\boldsymbol{0}\\ \boldsymbol{e}_t\end{bmatrix}, \qquad \|\boldsymbol{v}_{n+t}^{*}\|_2 = 1.$$

First n vectors of $\boldsymbol{B}^{'}$ are being called by q-vectors. For $\Lambda = \{\boldsymbol{x} \in \mathbb{Z}^{m+n+1} : (\boldsymbol{A}_{m\times n}^{\top}|\boldsymbol{I}_m| - \boldsymbol{b})\boldsymbol{x} \equiv \boldsymbol{0} \mod q\}$, we can also construct a basis like $\boldsymbol{B}^{'}$ having first m many q-vectors.

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